

PRODUCT NAME

BEMUNAT 25 (Bendamustine Hydrochloride 25mg Powder for Concentrate for Solution for Infusion)

BEMUNAT 100 (Bendamustine Hydrochloride 100mg Powder for Concentrate for Solution for Infusion)

DOSAGE FORMS AND STRENGTHS

BEMUNAT 25 contain Bendamustine Hydrochloride 25mg

BEMUNAT 100 contain Bendamustine Hydrochloride 100mg

DESCRIPTION

BEMUNAT 25 or 100 Powder before reconstitution:

White to off white lyophilized cake or powder in amber colored vial with double slotted plugs with blue color flip off seal.

After reconstitution with sterile water for injection: Clear, colorless to pale yellow colored solution

After further dilution with 0.9% sodium chloride solution: Clear, colorless solution

INDICATIONS

BEMUNAT 25 and 100 is indicated for monotherapy in patients with chronic lymphocytic leukaemia. Efficacy relative to first line therapies other than chlorambucil has not been established.

BEMUNAT 25 and 100 is indicated for monotherapy in patients with indolent B-cell non-Hodgkin's lymphomas that has progressed during or within six months of treatment with rituximab or a rituximab-containing regimen.

DOSAGE AND ADMINISTRATION

For intravenous infusion over 30 - 60 minutes.

Infusion must be administered under the supervision of a physician qualified and experienced in the use of chemotherapeutic agents.

Poor bone marrow function is related to increased chemotherapy-induced haematological toxicity. Treatment should not be started if leukocyte and/or platelet values have dropped to $< 3,000/\mu\text{l}$ or $< 75,000/\mu\text{l}$, respectively.

Monotherapy for chronic lymphocytic leukaemia

100 mg/m^2 body surface area bendamustine hydrochloride on days 1 and 2; every 4 weeks.

Monotherapy for indolent non-Hodgkin's lymphomas refractory to rituximab

120 mg/m^2 body surface area bendamustine hydrochloride on days 1 and 2; every 3 weeks.

Treatment should be terminated or delayed if leukocyte and/or platelet values drop to $< 3000/\mu\text{L}$ or $< 75000/\mu\text{L}$, respectively. Treatment can be continued after leukocyte values have increased to $> 4000/\mu\text{L}$ and platelet values to $> 100000/\mu\text{L}$.

The leukocyte and platelet nadir is reached after 14-20 days with regeneration after 3-5 weeks. During therapy free intervals strict monitoring of the blood count is recommended.

In case of non-hematological toxicity, dose reductions have to be based on the worst common toxicity criteria (CTC) grades in the preceding cycle. A 50% dose reduction is recommended in case of CTC grade 3 toxicity. An interruption of treatment is recommended in case of CTC grade 4 toxicity.

If a patient requires a dose modification, the individually calculated reduced dose must be given on day 1 and 2 of the respective treatment cycle.

Hepatic impairment

On the basis of pharmacokinetic data, no dose adjustment is necessary in patients with mild hepatic impairment (serum bilirubin < 1.2 mg/dL). A 30% dose reduction is recommended in patients with moderate hepatic impairment (serum bilirubin 1.2 - 3.0 mg/dL).

No data are available in patients with severe hepatic impairment (serum bilirubin values of > 3.0 mg/dL).

Renal impairment

On the basis of pharmacokinetic data, no dose adjustment is necessary in patients with a creatinine clearance of > 10 mL/min. Experience in patients with severe renal impairment is limited.

Pediatric patients

As there are limited data, the safety and efficacy of bendamustine in pediatric patients has not been established.

Elderly patients

There is no evidence that dose adjustments are necessary in elderly patients.

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients (see List of Excipients)
- During breast-feeding
- Severe hepatic impairment (serum bilirubin > 3.0 mg/dL)
- Jaundice
- Severe bone marrow suppression and severe blood count alterations (leukocyte and/or platelet values dropped to < 3000/ μ L or < 75000/ μ L, respectively)
- Major surgery less than 30 days before start of treatment
- Infections, especially involving leukocytopenia
- Yellow fever vaccination

WARNINGS AND PRECAUTIONS

Myelosuppression

Patients treated with bendamustine hydrochloride may experience myelosuppression (bone marrow failure). In the event of treatment-related myelosuppression, leukocytes, platelets, hemoglobin, and neutrophils should be monitored and re-evaluated prior to initiation of

the next cycle of therapy. Prior to the initiation of the next cycle of therapy, the following parameters are recommended:

Leukocyte and/or platelet values $> 4000/\mu\text{L}$ or $> 100000/\mu\text{L}$, respectively. Treatment-related myelosuppression may require dose adjustment and/or dose delays.

Treatment with bendamustine hydrochloride may cause prolonged lymphocytopenia ($< 600/\mu\text{L}$) and low CD4-positive T-cell (T-helper cell) counts ($< 200/\mu\text{L}$) for at least 7–9 months after the completion of treatment. Lymphocytopenia and CD4-positive T-cell depletion are more pronounced when bendamustine is combined with rituximab. Patients with lymphopenia and low CD4-positive T-cell count following treatment with bendamustine hydrochloride are more susceptible to (opportunistic) infections.

BEMUNAT should not be used during severe bone marrow suppression and severe blood count alterations.

Infections

Serious and fatal infections, including fatal sepsis, have occurred with bendamustine treatment. These infections included bacterial (pneumonia) and opportunistic infections such as Pneumocystis Jirovecii Pneumonia (PJP), Varicella Zoster Virus (VZV) and Cytomegalovirus (CMV). Cases of progressive multifocal leukoencephalopathy (PML) including fatal ones have been reported following the use of bendamustine mainly in combination with rituximab or obinutuzumab. Patients with lymphopenia and low CD4-positive T-cell count following treatment with bendamustine hydrochloride are more susceptible to (opportunistic) infections. In case of low CD4-positive T-cell counts ($< 200/\mu\text{L}$) Pneumocystis jirovecii pneumonia (PJP) prophylaxis should be considered. All patients should be monitored for respiratory signs and symptoms throughout treatment. Discontinuation of bendamustine hydrochloride should be considered if there are signs of (opportunistic) infections. Consider PML in the differential diagnosis in patients with new or worsening neurological, cognitive or behavioural signs or symptoms. If PML is suspected then appropriate evaluations should be undertaken and treatment suspended until PML is excluded.

Skin reactions

A number of skin reactions have been reported. These events have included rash, toxic skin reactions and bullous exanthema. Cases of Stevens – Johnson syndrome (SJS) and Toxic Epidermal Necrolysis (TEN), some fatal, have also been reported. Some events of SJS and TEN occurred when bendamustine hydrochloride was administered concomitantly with allopurinol or when bendamustine hydrochloride was given in combination with other anticancer agents. Cases of drug reaction with eosinophilia and systemic symptoms (DRESS) have been reported with the use of bendamustine hydrochloride in combination with rituximab. Where skin reactions occur, they may be progressive and increase in severity with further treatment; therefore, patients with skin reactions should be monitored closely. If skin reactions are severe or progressive, BEMUNAT should be withheld or discontinued. For severe skin reactions where a relationship to bendamustine hydrochloride is suspected, treatment should be discontinued.

Patients with cardiac disorders

During treatment with bendamustine hydrochloride the concentration of potassium in the blood should be closely monitored. Potassium supplementation should be given when $\text{K}^+ < 3.5 \text{ mEq/L}$, and ECG measurement must be performed.

Fatal cases of myocardial infarction and cardiac failure have been reported with bendamustine treatment. Patients with concurrent or history of cardiac disease should be observed closely.

Nausea, vomiting

An antiemetic may be given for the symptomatic treatment of nausea and vomiting.

Tumor lysis syndrome

Tumor lysis syndrome associated with BEMUNAT treatment has been reported in patients in clinical trials. The onset tends to be within 48 hours of the first dose of BEMUNAT and, without intervention, may lead to acute renal failure and death. Preventive measures include adequate volume status, close monitoring of blood chemistry, particularly potassium and uric acid levels. The use of allopurinol during the first one to two weeks of BEMUNAT therapy can be considered but not necessarily as standard.

Anaphylaxis

Infusion reactions to bendamustine hydrochloride have occurred commonly in clinical trials.

Symptoms are generally mild and include fever, chills, pruritus and rash. In rare instances severe anaphylactic and anaphylactoid reactions have occurred. Patients must be asked about symptoms suggestive of infusion reactions after their first cycle of therapy. Measures to prevent severe reactions, including administration of antihistamines, antipyretics and corticosteroids must be considered in subsequent cycles in patients who have previously experienced infusion reactions.

Patients who experienced Grade 3 or worse allergic-type reactions were typically not re- challenged.

Contraception

Bendamustine hydrochloride is teratogenic and mutagenic.

Women should not become pregnant during treatment. Male patients should not father a child during and up to 6 months after treatment. They should seek advice about sperm conservation prior to treatment with BEMUNAT because of possible irreversible infertility.

Extravasation

An extravasal injection should be stopped immediately. The needle should be removed after a short aspiration. Thereafter the affected area of tissue should be cooled. The arm should be elevated. Additional treatments, such as the use of corticosteroids are not of clear benefit.

Non-melanoma skin cancer

In clinical studies, an increased risk for non-melanoma skin cancers (basal cell carcinoma and squamous cell carcinoma) has been observed in patients treated with bendamustine containing therapies. Periodic skin examination is recommended for all patients, particularly those with risk factors for skin cancer.

Other malignancies

There are reports of secondary tumors, including myelodysplastic syndrome, myeloproliferative disorders, acute myeloid leukemia and bronchial carcinoma. The association with bendamustine therapy has not been determined.

Hepatitis B reactivation

Reactivation of hepatitis B in patients who are chronic carriers of this virus has occurred after these patients received bendamustine hydrochloride. Some cases resulted in acute hepatic failure or a fatal outcome. Patients should be tested for HBV infection before initiating treatment with bendamustine hydrochloride. Carriers of HBV who require treatment with bendamustine hydrochloride should be closely monitored for signs and symptoms of active HBV infection throughout therapy and for several months following termination of therapy.

INTERACTIONS

No *in-vivo* interaction studies have been performed.

When BEMUNAT is combined with myelosuppressive agents, the effect of BEMUNAT and/or the co-administered medicinal products on the bone marrow may be potentiated. Any treatment reducing the patient's performance status or impairing bone marrow function can increase the toxicity of BEMUNAT.

Combination of BEMUNAT with cyclosporine or tacrolimus may result in excessive immunosuppression with risk of lymphoproliferation.

Cytostatics can reduce antibody formation following live-virus vaccination and increase the risk of infection which may lead to fatal outcome. This risk is increased in subjects who are already immunosuppressed by their underlying disease.

Bendamustine metabolism involves cytochrome P450 (CYP) 1A2 isoenzyme. Therefore, potential for interaction with CYP1A2 inhibitors (such as fluvoxamine, ciprofloxacin, acyclovir, cimetidine) exists.

PREGNANCY, BREAST-FEEDING AND FERTILITY

Pregnancy

There are insufficient data from the use of BEMUNAT in pregnant women. In non-clinical studies bendamustine was embryo-/feto-lethal, teratogenic and genotoxic. Women of childbearing potential must use effective methods of contraception both before, during, and one month following BEMUNAT therapy.

During pregnancy BEMUNAT should not be used unless the benefit outweighs the risk. The mother should be informed about the risk to the fetus.

If treatment with BEMUNAT is absolutely necessary during pregnancy or if pregnancy occurs during treatment, the patient should be informed about the risks for the unborn child and be monitored carefully. The possibility of genetic counseling should be considered.

Breast-feeding

It is not known whether bendamustine passes into the breast milk, therefore,

BEMUNAT is contraindicated during breast-feeding. Breast-feeding must be discontinued during treatment with BEMUNAT.

Fertility

Men being treated with bendamustine are advised not to father a child during and for up to 6 months following cessation of treatment. Advice on conservation of sperm should be sought prior to treatment because of the possibility of irreversible infertility due to therapy with BEMUNAT.

Effects on Ability to Drive and Use Machines

No studies on the effects on the ability to drive and use machines have been performed. However, ataxia, peripheral neuropathy and somnolence have been reported during treatment with Bendamustine. Patients should be instructed that if they experience these symptoms they should avoid potentially hazardous tasks such as driving and using machines.

ADVERSE REACTIONS

The most common adverse reactions with bendamustine hydrochloride are hematological adverse reactions (leukopenia, thrombopenia), dermatologic toxicities (allergic reactions), constitutional symptoms (fever), gastrointestinal symptoms (nausea, vomiting).

The table below reflects the data obtained with bendamustine hydrochloride in clinical trials.

MedDRA system organ class	Very common ≥1/10	Common ≥1/100 to <1/10	Uncommon ≥1/1000 to <1/100	Rare ≥1/10000 to <1/1000	Very rare <1/10000	Not known (cannot be estimated from the available data)
Infections and Infestations	Infection NOS*			Sepsis	Pneumonia primary atypical	
Neoplasm benign, malignant		Tumor lysis syndrome				
Blood and lymphatic system disorders	Leukopenia NOS*, Thrombocytopenia	Hemorrhage, Anemia, Neutropenia			Hemolysis	
Immune system disorders		Hypersensitivity NOS*		Anaphylactic reaction, Anaphylactoid reaction	Anaphylactic shock	

Nervous system disorders		Insomnia		Somnolence, Aponia	Dysgeusia, Paresthesia, Peripheral sensory neuropathy, Anticholinergic syndrome, Neurological disorders, Ataxia, Encephalitis	
Cardiac disorders		Cardiac dysfunction, such as palpitations, angina pectoris, Atrial fibrillation	Pericardial effusion		Tachycardia, Myocardial infarction, Cardiac failure	
Vascular disorders		Hypotension		Acute circulatory failure	Phlebitis	
Respiratory, thoracic and mediastinal disorders		Pulmonary dysfunction			Pulmonary fibrosis	
Gastrointestinal disorders	Nausea, vomiting	Diarrhea, Constipation, Stomatitis				Hemorrhagic esophagitis, Gastrointestinal hemorrhage
Skin and subcutaneous tissue disorders		Alopecia, Skin disorders NOS*		Erythema, Dermatitis, Pruritus, Maculopapular rash, Hyperhidrosis		
Reproductive system and breast disorders		Amenorrhea			Infertility	
General disorders and administration site conditions	Mucosal inflammation, Fatigue, Pyrexia	Pain, Chills, Dehydration, Anorexia			Multi organ failure	
Investigations	Hemoglobin decrease, Creatinine increase, Urea increase	AST increase, ALT increase, Alkaline phosphatase increase, Bilirubin increase Hypokalemia				

(NOS* = Not otherwise specified)

A small number of cases of Stevens-Johnson syndrome and toxic epidermal necrolysis have been reported in patients some using bendamustine in combination with allopurinol or in combination with allopurinol and rituximab. In addition, a few cases of hepatitis B reactivation resulting in hepatic failure have been reported in patients treated with bendamustine. Pancytopenia, headache, dizziness, opportunistic infection (e.g. herpes zoster, cytomegalovirus, pneumocystis jirovecii pneumonia), bone marrow failure, hepatic failure, renal failure, nephrogenic diabetes insipidus, drug reaction with eosinophilia and systemic symptoms (combination therapy with rituximab) have also been reported in patients treated with bendamustine.

The CD4/CD8 ratio may be reduced. A reduction of the lymphocyte count was seen. In immunosuppressed patients, the risk of infection (e.g., with herpes zoster) may be increased.

There have been isolated reports of necrosis after accidental extra-vascular administration, tumor lysis syndrome, and anaphylaxis.

The risk of myelodysplastic syndrome and acute myeloid leukaemias is increased in patients treated with alkylating agents (including bendamustine). The secondary malignancy may develop several years after chemotherapy has been discontinued.

OVERDOSE

After application of a 30-minute infusion of bendamustine once every 3 weeks, the maximum tolerated dose (MTD) was 280 mg/m². Cardiac events of CTC grade 2 occurred, which were compatible with ischemic ECG changes that were regarded as dose limiting.

In a subsequent study with a 30-minute infusion of bendamustine at day 1 and 2 every 3 weeks, the MTD was found to be 180 mg/m². The dose limiting toxicity was grade 4 thrombocytopenia. Cardiac toxicity was not dose limiting with this schedule.

Counter measures

There is no specific antidote. Bone marrow transplantation and transfusions (platelets, concentrated erythrocytes) may be made or hematological growth factors may be given as effective countermeasures to control hematological side-effects.

Bendamustine hydrochloride and its metabolites are dialyzable to a small extent.

PHARMACOLOGICAL PROPERTIES

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Antineoplastic agents, alkylating agents, ATC code: L01AA09

Bendamustine is an alkylating antitumor agent with unique activity containing a purine-like benzimidazole ring. The antineoplastic and cytotoxic effect of bendamustine is based essentially on a cross-linking of DNA single and double strands by alkylation. As a result, DNA matrix functions and DNA synthesis and repair are impaired. Bendamustine is active

against both quiescent and dividing cells. The exact mechanism of action of bendamustine remains unknown.

The antitumor effect of bendamustine hydrochloride has been demonstrated by several *in vitro* studies in different human tumor cell lines (breast cancer, non-small cell and small cell lung cancer, ovary carcinoma and different leukemia) and *in vivo* in different experimental tumor models with tumors of mouse, rat and human origin (melanoma, breast cancer, sarcoma, lymphoma, leukemia and small cell lung cancer).

Bendamustine hydrochloride showed an activity profile in human tumor cell lines different to that of other alkylating agents. The active substance revealed no or very low cross-resistance in human tumor cell lines with different resistance mechanisms at least in part due to a comparatively persistent DNA interaction.

PHARMACOKINETIC PROPERTIES

Distribution

Following 30 min i.v. infusion the central volume of distribution was 19.3 l. Under steady-state conditions following i.v. bolus injection the volume of distribution was 15.8- 20.5 l. More than 95% of the substance is bound to plasma proteins (primarily albumin).

Metabolism

A major route of clearance of bendamustine is the hydrolysis to monohydroxy- and dihydroxy-bendamustine. Formation of N-desmethyl-bendamustine and gamma-hydroxybendamustine by hepatic metabolism involves cytochrome P450 (CYP) 1A2 isoenzyme. Another major route of bendamustine metabolism involves conjugation with glutathione. In-vitro bendamustine does not inhibit CYP 1A4, CYP 2C9/10, CYP 2D6, CYP 2E1 or CYP 3A4.

Elimination

About 20 % of the administered dose was recovered in urine within 24 hours. Amounts excreted in urine were in the order monohydroxy-bendamustine > bendamustine > dihydroxy-bendamustine > oxidised metabolite > N-desmethyl bendamustine. In the bile, primarily polar metabolites are eliminated.

Hepatic impairment

There was no significant difference to patients with normal liver and kidney function with respect to C_{max}, t_{max}, AUC, t_{1/2β}, volume of distribution and clearance. AUC and total body clearance of bendamustine correlate inversely with serum bilirubin.

Renal impairment

In patients with creatinine clearance > 10 ml/min including dialysis dependent patients, no significant difference to patients with normal liver and kidney function was observed with respect to C_{max}, t_{max}, AUC, t_{1/2β}, volume of distribution and clearance.

Elder subjects

Higher age does not influence the pharmacokinetics of bendamustine.

PHARMACEUTICAL

PARTICULARS List of Excipients

Mannitol

INCOMPATIBILITIES

This medicinal product should not be mixed with other medicinal products except those mentioned in Instructions for Use and Handling and Disposal.

SHELF LIFE

Unopened vial: 24 months.

After Opening:

The powder should be reconstituted immediately with water for injection (WFI) after opening of the vial.

The reconstituted concentrate should be diluted immediately with 0.9% w/v Sodium chloride solution.

Solution for infusion

After reconstitution and dilution, the solution should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

STORAGE CONDITIONS

Store below 30°C in original packaging.

Keep the vial in the outer carton in order to protect from light.

For storage conditions of the reconstituted or diluted medicinal product, see section shelf life.

Keep out of sight and reach of children.

NATURE AND CONTENTS OF CONTAINER

Bendamustine is provided in type I amber glass vials with 20mm double slotted rubber stopper and 20mm blue flip-off seal.

20 mL-vials contain 25 mg bendamustine hydrochloride and are supplied in packs of 1 vial.

50 mL-vials contain 100 mg bendamustine hydrochloride and are supplied in packs of 1 vial.

INSTRUCTIONS FOR USE AND HANDLING

When handling BEMUNAT, inhalation, skin contact or contact with mucous membranes should be avoided (wear gloves and protective clothes!). Contaminated body parts should be carefully rinsed with water and soap, the eye should be rinsed with physiological saline solution. If possible, it is recommended to work on special safety workbenches (laminar flow) with liquid impermeable, absorbing disposable foil. Pregnant personnel should be excluded from handling cytostatics

The powder for concentrate for solution for infusion has to be reconstituted with water for injection, diluted with sodium chloride 9 mg/mL (0.9%) solution for injection and then administered by intravenous infusion. Aseptic technique is to be used.

1. Reconstitution

Reconstitute each vial of BEMUNAT containing 25 mg bendamustine hydrochloride in 10 mL water for injection by shaking.

Reconstitute each vial of BEMUNAT containing 100 mg bendamustine hydrochloride in 40 mL water for injection by shaking.

The reconstituted concentrate contains 2.5 mg bendamustine hydrochloride per ml and appears as a clear, colorless to pale yellow colored solution

2. *Dilution*

As soon as a clear solution is obtained (usually after 5-10 minutes) dilute the total recommended dose of BEMUNAT immediately with 0.9% Sodium chloride solution to produce a final volume of about 500 mL.

BEMUNAT must be diluted with 0.9% Sodium chloride solution and not with any other injectable solution.

3. *Administration*

The solution is administered by intravenous infusion over 30-60 min. The vials are for single use only. Infusion must be administered under the supervision of a physician qualified and experienced in the use of chemotherapy agents.

INSTRUCTIONS FOR DISPOSAL

Any unused product or waste material should be disposed of in accordance with local requirements.

MANUFACTURER

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PRODUCT REGISTRATION HOLDER

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